

OPERATION BLACK FALCON

Computer Forensics Investigation Assignment

Case ID: HIVE-CF-2026-007

Evidence ID: HIVE-EVID-001

Case Type: Computer Forensics / Extremism-Related Digital Investigation

Primary Subject: Osama Elorm Bin-Laden

Device: Seized Windows Laptop

Expected Hostname: To be determined by the examiner

Time Zone: GMT / Africa-Accra

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Training Notice: This is a completely fictional digital-forensics training scenario. All people, communications, transactions, locations, documents and events contained in the dataset are synthetic and created solely for forensic training.

Scenario

A national security agency has seized a laptop during a lawful investigation into the activities of a suspected extremist network operating through a combination of online communications, cloud-storage platforms, physical meetings and financial intermediaries.

The laptop is believed to belong primarily to a suspected facilitator known as **Osama Elorm Bin-Laden**.

Intelligence received before the seizure suggested that the subject may have been involved in coordinating individuals, maintaining contact databases, communicating with associates, managing financial contributions, distributing digital material, arranging meetings and transferring information using removable storage devices and cloud services.

Several individuals have emerged during the investigation as potentially relevant to the subject's activities:

- **Osama Elorm Bin-Laden** — primary subject
- **Nancy Aggrey** — person of investigative interest
- **Mary Gyasi** — person of investigative interest
- **Paul Addo** — person of investigative interest
- **Philip Adanse** — person whose relationship with the subject requires examination

Investigators believe the seized laptop may contain evidence relating to:

- communications between members of the network;
- financial transactions and contribution records;
- contact and membership lists;
- meeting arrangements;
- cloud-storage activity;
- browser activity and searches;
- documents and media files;
- removable USB storage;
- encrypted archives or volumes;
- deleted information;
- attempts to conceal or destroy evidence;
- geographic information;
- relationships between persons of interest.

The device has been forensically acquired and provided to you for examination.

Your responsibility is to conduct an independent forensic examination and determine what the digital evidence actually supports.

You must not assume that every person appearing in the evidence is involved in criminal activity. Your conclusions must be based strictly on recoverable forensic artifacts.

Student Instructions

Conduct the examination using an appropriate forensic platform such as:

- Autopsy
- FTK Imager
- Registry Explorer
- DB Browser for SQLite
- Timeline Explorer
- Thunderbird-compatible forensic tools
- Other approved forensic utilities

For every significant finding, record where applicable:

- artifact type;
- source path;
- filename;
- relevant account;
- timestamp;

- time zone;
- associated person;
- hash value where relevant;
- supporting forensic artifact.

Do not rely on a single artifact where corroboration is possible.

PART 1 — Evidence Validation and Acquisition

Question 1

Identify the forensic evidence provided for examination and document:

- evidence filename;
- evidence format;
- evidence size;
- acquisition identifier;
- case number;
- evidence number.

Question 2

Calculate and record the available cryptographic hash values of the forensic image.

Do the calculated values match the supplied acquisition or verification records?

Question 3

What chain-of-custody information is available for the evidence?

Identify:

- examiner;
- organization;
- acquisition date;
- evidence reference;
- acquisition notes;
- ownership information.

Question 4

Is there any evidence within the forensic dataset identifying its creator or owner?

Document every location in which ownership or attribution information is found.

PART 2 — System Identification

Question 5

Determine the computer's:

- hostname;
- operating system;
- Windows edition;
- operating-system version;
- build number;
- architecture;
- configured time zone.

Identify the forensic artifacts used to support your findings.

Question 6

Identify all user accounts present on the system.

For each account determine, where possible:

- username;
- profile path;
- account type;
- last logon;
- whether the account appears active or dormant.

Which account appears to have been the primary user of the laptop?

Question 7

Determine when the computer was:

- last successfully started;
- last logged into;
- last normally shut down;

- last unexpectedly powered off.

Explain the artifacts used to determine these events.

PART 3 — Installed Applications and Program Execution

Question 8

Identify significant applications installed on the laptop.

Pay particular attention to applications associated with:

- browsers;
- messaging;
- cloud storage;
- encryption;
- password management;
- compression;
- secure deletion;
- system cleaning.

Question 9

Which applications appear to have been used most frequently?

Support your answer using execution artifacts such as:

- Prefetch;
- UserAssist;
- Amcache;
- application databases;
- Jump Lists;
- recent-file artifacts;
- event logs.

Question 10

Is there evidence that cleaning, privacy or anti-forensics software was installed or executed?

If so, identify:

- application;
- execution time;
- affected user;
- supporting artifact.

PART 4 — USB and External Devices

Question 11

Identify all removable or external devices that were connected to the laptop.

For each device determine, where possible:

- manufacturer;
- device name;
- serial number;
- volume label;
- first connection;
- last connection;
- associated user.

Question 12

Was any USB storage device used during the final seven days of activity?

If so, identify files that may have been:

- opened from the device;
- copied to the laptop;
- copied from the laptop;
- deleted from the removable device.

Question 13

Is there evidence connecting **Paul Addo** to any USB-related activity, logistics record or transferred file?

Present all relevant evidence.

PART 5 — Browser Forensics

Question 14

Identify all web browsers used by the primary user.

Locate the principal browser databases and record their forensic paths.

Question 15

What websites were visited during the final 30 days before the laptop was seized?

Group the significant websites into categories such as:

- webmail;
- cloud storage;
- search engines;
- mapping;
- social media;
- file sharing;
- general browsing.

Question 16

Identify significant search queries entered into the browsers by the primary user.

For each relevant search record:

- query;
- browser;
- timestamp;
- source database.

Question 17

Identify files downloaded through the browsers.

For significant downloads determine:

- filename;
- source URL or domain;
- download time;
- destination path;
- associated browser.

Question 18

Is there evidence that browser history, cookies, downloads or recent activity were intentionally removed?

Explain your findings.

PART 6 — Email Examination

Question 19

Identify all email accounts or email profiles present on the system.

Determine whether email activity occurred through:

- a dedicated email client;
- browser-based webmail;
- exported email files.

Question 20

Identify significant email communications involving:

- Nancy Aggrey;
- Mary Gyasi;
- Paul Addo;
- Philip Adanse.

For each significant communication record:

- sender;

- recipient;
- date/time;
- subject;
- attachment where applicable;
- relevance to the investigation.

PART 7 — Messaging and Communications

Question 21

Identify messaging platforms or communication databases found on the laptop.

Question 22

Examine the communications database and determine:

- total number of messages;
- most frequent contacts;
- most frequent sender;
- most frequent recipient;
- communication frequency by person.

Question 23

Identify the principal phone numbers associated with the persons of interest.

Do all relevant training phone numbers follow the expected numbering pattern?

Question 24

Determine which of the following individuals communicated most frequently with the primary subject:

- Nancy Aggrey;
- Mary Gyasi;
- Paul Addo;
- Philip Adanse.

Support your answer quantitatively.

Question 25

Can communications between the subjects be represented through link analysis?

Construct a relationship diagram showing:

- nodes;
- communication frequency;
- direction of communication;
- important shared files or events.

PART 8 — Documents and Office Files

Question 26

Identify documents, spreadsheets, PDFs and presentations relevant to the investigation.

Document:

- filename;
- file type;
- path;
- author;
- created date;
- modified date;
- last accessed date where available;
- investigative relevance.

Question 27

Examine Office document metadata.

Are any documents associated with:

- Osama Elorm Bin-Laden;
- Nancy Aggrey;
- Mary Gyasi;
- Paul Addo;
- Philip Adanse?

Question 28

Identify documents relating to:

- contact lists;
- finances;
- meetings;
- logistics;
- communications;
- cloud sharing;
- membership or registration.

Explain how the documents relate to other forensic artifacts.

PART 9 — Images and Media Evidence

Question 29

Identify the significant images stored on the laptop.

For each relevant image determine:

- filename;
- path;
- file type;
- dimensions;
- metadata;
- creation or modification date;
- embedded ownership information.

Question 30

Determine whether any images can be associated with particular persons of interest.

Do not identify individuals solely from facial appearance. Base the association on filenames, surrounding documents, metadata, messages, IDs or other forensic evidence.

Question 31

Are there duplicate, resized, converted, cached or thumbnail versions of relevant images?

If so, determine whether the copies can be linked to the same original file.

PART 10 — Financial Investigation

Question 32

Identify financial records present on the system.

Examine:

- spreadsheets;
- PDFs;
- transaction records;
- receipts;
- communications;
- cloud files.

Question 33

Is **Mary Gyasi** associated with any financial activity?

If so, identify the transactions, documents and communications supporting the association.

Question 34

Can financial records be corroborated using more than one artifact source?

For example, determine whether a transaction appears in:

- a spreadsheet;
- a PDF receipt;
- a message;
- an email;
- a cloud record.



PART 11 — Cloud Forensics

Question 35

Was cloud storage used on the laptop?

Investigate potential evidence relating to:

- Google Drive;
- OneDrive;
- Dropbox.

Question 36

For each cloud platform identified, determine:

- associated account;
- files uploaded;
- files downloaded;
- shared files;
- synchronization activity;
- timestamps.

Question 37

Identify any files that appear both on the local computer and in cloud-storage records.

What does this indicate about file movement?

PART 12 — Encryption and Password Evidence

Question 38

Identify any encrypted:

- archives;
- containers;
- volumes;
- password databases.

Question 39

Is there evidence of a password manager on the system?

If so, determine:

- application;
- database location;
- associated user;
- relevant metadata.

Question 40

Are there encrypted-drive artifacts present?

Document:

- volume/container name;
- apparent encryption type;
- associated files;
- recovery information if discoverable.

Do not attempt to bypass encryption beyond the scope authorized by the exercise.

Question 41

Can any recovery keys, passwords or password hints be located elsewhere in the forensic image?

If so, document the exact artifact and its relationship to the encrypted evidence.

PART 13 — Deleted and Recoverable Evidence

Question 42

Identify deleted files that can be recovered.

For each significant recovered file record:

- original filename;
- original location;
- deletion status;

- recovered content;
- timestamp;
- investigative significance.

Question 43

Do deleted files contain evidence that also appears elsewhere on the system?

Correlate recovered material with:

- browser history;
- communications;
- financial records;
- cloud activity;
- USB evidence.

PART 14 — Geolocation Examination

Question 44

Identify all available geolocation evidence.

Possible sources may include:

- EXIF information;
- KML files;
- mapping searches;
- cloud records;
- Wi-Fi data;
- documents;
- messages.

Question 45

What locations appear most frequently in the evidence?

For each significant location identify the artifacts supporting it.

Question 46

Can any location be correlated with a communication, document, USB event or meeting record?

Construct at least one multi-artifact correlation.

PART 15 — Anti-Forensics and Concealment

Question 47

Identify evidence that the primary user attempted to conceal activity.

Investigate:

- deleted files;
- browser clearing;
- privacy utilities;
- file renaming;
- encryption;
- hidden files;
- archive creation;
- metadata manipulation.



Question 48

Is there evidence of timestamp manipulation or unusual file-time inconsistencies?

Identify any suspicious examples and explain why they require further investigation.



PART 16 — Final Seven-Day Timeline

Question 49

Construct a forensic timeline covering the final **seven days before seizure**.

Your timeline should incorporate, where available:

- logon activity;
- browser activity;
- USB connections;
- document access;
- communications;
- financial records;
- cloud activity;
- encryption;

- deletions;
- anti-forensics;
- final power event.

Question 50

Identify the **ten most significant events** occurring during those seven days.

For each event provide:

1. date/time;
2. activity;
3. user;
4. artifact;
5. source path;
6. investigative significance.

PART 17 — Relationship Analysis

Question 51

Determine the relationship between the primary user and each of the following:

Nancy Aggrey

What evidence establishes the relationship?

Mary Gyasi

What evidence establishes the relationship?

Paul Addo

What evidence establishes the relationship?

Philip Adanse

What evidence establishes the relationship?

Question 52

Which three individuals appear most closely connected to the subject based on the totality of digital evidence?

Use evidence rather than assumptions.

PART 18 — Intelligence Corroboration

Question 53

Does the forensic evidence corroborate the intelligence allegation that the laptop was being used for coordinated communications?

Explain.

Question 54

Does the evidence support the allegation that financial records were maintained on the laptop?

Explain.

Question 55

Does the evidence support the allegation that cloud storage was used to move or share information?

Explain.

Question 56

Does the evidence support the allegation that removable USB storage was used?

Explain.

Question 57

Does the evidence demonstrate attempts to conceal, delete or encrypt information?

Explain.

Question 58

Identify any intelligence allegation that is **not sufficiently supported** by the digital evidence.

Explain why the evidence is insufficient.

PART 19 — Final Investigative Assessment

Question 59

Based solely on the digital evidence, describe the primary user's activities on the seized laptop.

Your answer must distinguish between:

- established fact;
- reasonable forensic inference;
- unsupported allegation.

Question 60

Prepare a final evidence-based conclusion addressing:

- identity of the primary user;
- key associates;
- communication activity;
- significant documents;
- financial evidence;
- USB activity;
- browser activity;
- cloud activity;
- encryption;
- deleted evidence;
- geolocation;
- concealment attempts;
- significant events immediately before seizure.

Conclude by identifying the **strongest five pieces of digital evidence** in the case and explain why each is important

Submission Requirement

Students should produce a professional forensic report containing:

Executive Summary → Evidence Handling → Examination Methodology → Findings → Timeline → Relationship Analysis → Supporting Screenshots → Conclusions → Appendices

Every major conclusion should be supported by a forensic artifact and timestamp.

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